

Stack Applications: Well-Formedness of Parentheses

Q. Explain how a stack can be used to check the well-formedness of parentheses in an expression. Write a step-by-step algorithmic approach.

> **Definition to Remember**

> An opening bracket must be followed by its closing

1. Why Stack Works for This Problem

- LIFO ensures the **most recently opened bracket** is at the top of the stack.

- When an opening bracket is encountered, it is pushed onto the stack. When a closing bracket is encountered, it is popped from the stack and compared with the opening bracket.

2. Algorithm

```

1. Initialize an empty stack.

## 3. Trace Example: `[( )]`

| Character | Action | Stack State |
|-----------|--------|-------------|
|-----------|--------|-------------|

| Character | Action | Stack State |
|-----------|--------|-------------|
|-----------|--------|-------------|

**Invalid Example:** `{ ( ) }`

- Reason: The closing bracket `)` does not match the opening bracket `{`.

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> **Must-Write Points (for 10 marks)**

> 1. Stack is empty initially.

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> **Quick Recall**

> `Ope  
O(n)`

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## C Code: Parentheses Checking Algorithm using Stack

```
```c
```

```
#includ
```

```
#define MAX 100
```

```
struct CharStack {
```

```
char ite
```

```
void push(struct CharStack *s, char c) {
```

```
s->iter
```

```
char pop(struct CharStack *s) {
```

```
if (s->t
```

```
int isMatchingPair(char char1, char char2) {
```

```
if (char
```

```
int checkBalanced(char exp[]) {
```

```
struct t
```

```
for (int i = 0; i < strlen(exp); i++) {
```

```
if (expl
```

```
int main() {
```

```
char e
```

```
printf("Expression: %s -> %s\n", expr1, checkBalanced(expr1) ? "Balanced " : "Unbalanced ");
```

```
printf("
```

```
return 0;
```

```
}
```